

Morning Cheat Sheet — read this and nothing else

Frontend Developer · WeMakeScholars · 07:20-08:00 · Learn nothing new today — revision only.

07:20 – 08:00. Nothing else. No new material.

If a line here doesn't ring a bell, open that file for 3 minutes and come back.

The 8 sentences that carry this interview

1. **Why Next.js:** "Plain React ships an empty HTML shell that the browser fills in — bad for first paint and unreliable for crawlers. Next renders on the server, so the browser and Google both get real HTML immediately."
2. **Rendering strategies:** "SSG at build, SSR per request, ISR is SSG that regenerates on a timer, CSR in the browser. Bank listing pages → ISR. A student's own application page → SSR or CSR behind auth."
3. **Server vs Client Components:** "Server by default, ships zero JS; push `'use client'` down to the interactive leaves."
4. **Core Web Vitals:** "LCP under 2.5 s — is it there. CLS under 0.1 — does it stay still. INP under 200 ms — does it respond."
5. **Closure:** "A function that remembers the scope it was created in. It's how `useState` and `debounce` both work."
6. **Keys:** "Stable identity across renders. Index breaks the moment you delete or reorder — React reuses the wrong DOM node."
7. **Four UI states:** "Loading, error, empty, success. Empty is the one people forget."
8. **Business tie-in (say once):** "Since students find you through Google, server rendering and load speed here are lead generation, not polish."

JavaScript

- `var` function-scoped · `let` / `const` block-scoped + TDZ · `const` ≠ immutable object
- Event loop: **microtasks (Promises) before macrotasks (setTimeout)** → 1 4 3 2
- `Promise.all` parallel · `allSettled` all results · `race` first settled · `any` first success
- Arrow functions take `this` from the enclosing scope and can't be rebound
- `??` only falls back on null/undefined; `||` falls back on any falsy (0 !)
- Spread = **shallow** copy · deep = `structuredClone`
- `fetch` **does not throw on 404/500** — check `res.ok`
- Debounce = after they stop (search) · Throttle = at most once per N ms (scroll)
- `typeof null === 'object'` · `NaN !== NaN` · `Array.isArray()`

React

- Re-render causes: own state · props · **parent re-rendered** · context value
- Reduce: `React.memo` → `useCallback` / `useMemo` → move state down → `children`
- Never mutate state — new reference or no re-render (`Object.is`)
- `setCount(c => c + 1)` when the new value depends on the old
- `useEffect` = sync with the outside world; cleanup `return`; don't use it for derived data
- `useMemo` value · `useCallback` function · `useRef` persists without re-rendering
- Hooks: top level only — React tracks them **by call order**
- Context re-renders every consumer; memoize the provider value
- `React.lazy` + `Suspense` = code splitting

Next.js

- `app/page.js` routing · `[id]` dynamic · `layout.js` · `loading.js` · `error.js`
- `fetch(url, { next: { revalidate: 60 } })` = ISR · `{ cache: 'no-store' }` = SSR
- `next/image`: WebP/AVIF, srcset, lazy, reserves space (CLS) · `priority` for the LCP image
- `metadata` / `generateMetadata` for per-page title & description
- Hydration = server HTML + client listeners; mismatches come from `Date`, `random`, `window`
- `<Link>` client-side nav + prefetch; `<a>` = full reload
- App Router → `next/navigation`; Pages Router → `next/router`
- Only `NEXT_PUBLIC_*` reaches the browser — never a secret

CSS

- `box-sizing: border-box` · Flex = 1D, Grid = 2D
- `grid-template-columns: repeat(auto-fit, minmax(260px, 1fr))` = responsive, no media queries
- Mobile-first with `min-width` · viewport meta tag or nothing works
- `display:grid; place-items:center` to centre
- Animate `transform` / `opacity` only — they skip layout and paint

APIs

- 401 = who are you · 403 = I know, still no · 429 = rate limited
- `AbortController` in the effect cleanup → no stale state, no race condition
- CORS is a browser rule, fixed on the **backend**

System design — the 6 step framework

Clarify → Components → State → Data → Edge cases → Performance

- **Never start without asking 2 questions.** "Client side or API filtering?" "How many items?" "Does this need to rank on Google?"
- **Autocomplete:** debounce 400ms · four states · cancel the old request with `AbortController` · cache per query.
- **Listing page:** keep filters **in the URL** so links are shareable, back works, and the server can render it. Reset to page 1 when a filter changes.
- **Multi step form:** all data lives in the **parent**, steps are display only. Otherwise going back destroys what was typed.
- **Reusable component:** props say *what* not *how* · use `children` · spread `...rest` · a component with 15 boolean flags should be split.

Machine coding — the 6 things they're grading

1. `key={item.id}`, never the index
2. Derived data computed during render, not stored in state + effect
3. Empty state handled ("No results for...")
4. Loading **and** error states, ideally with a retry

5. Cleanup — `clearTimeout` / `clearInterval` / `abort`

6. Talking while you type

Opening & closing lines

"**Tell me about yourself**" → Present → Path → **Why here** (file 09 §1). Under 2 min.

Your closing question → *"What's the biggest frontend problem you're hoping this hire helps solve?"*

Filter questions — answer instantly, no hesitation:

18-month commitment ✓ · WFO ✓ · 1st & 3rd Saturdays ✓ · 10–11 AM login ✓

Salary → *"I saw the 4–6 LPA band and I'm comfortable in it; I'd expect to be placed based on how the technical rounds go."*

Never bid below 4.

The last 5 minutes before you walk in

- Close every tab except your GitHub and your two running projects.
- You know more than you think you do. Nobody expects perfection at this level — they expect someone who reasons clearly and doesn't bluff.
- **If you don't know something: say so, then reason out loud.** That answer has never lost anyone an offer. Bluffing has.
- Sit up. Breathe out slowly. Smile before you speak — it changes your voice.

Go get it.